Obeth Ovando Figueroa
AI

Assignment 2

For this assignment, I am working with the Frozen Lake code from Gymnasium.

The boundaries are represented using an 8x8 frozen lake with holes on the surface.
Our character's goal is to travel from the top left corner of the board to the bottom right corner of the board using Q-learning. Our character has limited movement to only move in cardinal directions (up, left, down, right). The episode ends when the character falls into a hole in the ice or the episode times itself out, avoiding an endless episode even if the possibilities of one is very slim.
The model goes through 10000 episodes to train and figure out the route from start to finish, then it takes the final Q-value table and runs the best route according to its learning.
The base environment I used had the following:

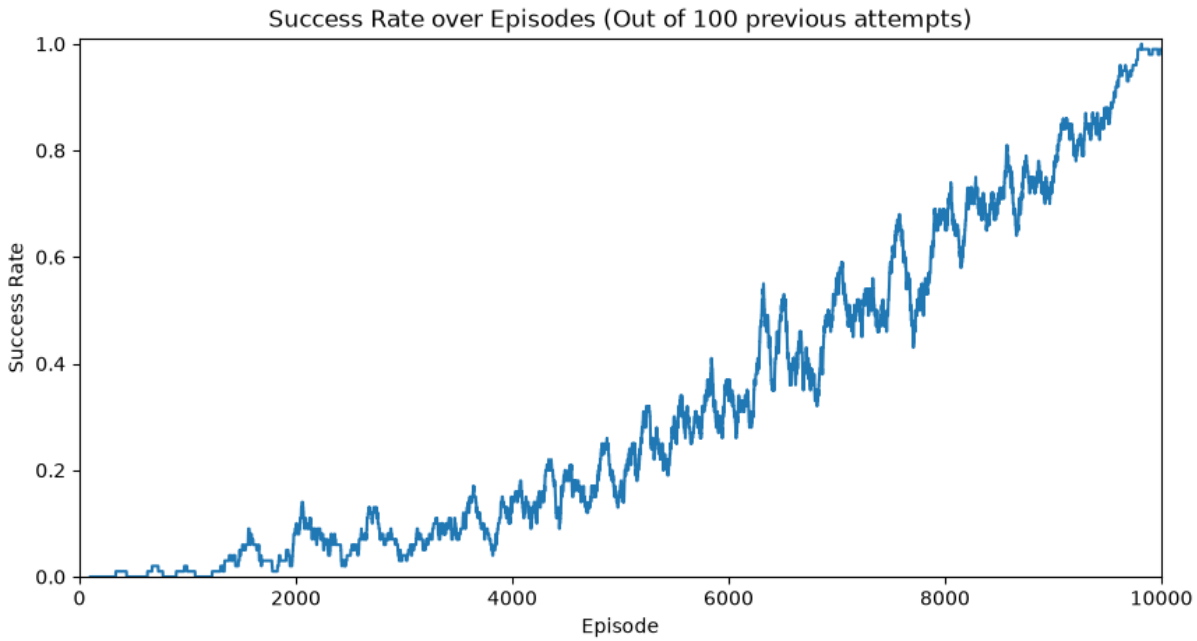
- **10000** Episodes
- A learning rate alpha of **0.5**
- A discount factor gamma of **0.9**
- An epsilon of **1**
- An epsilon decay of **0.0001**
- Time out limit of **500** steps
- A cost of **0.1** per step
- A reward of **10** when reaching the goal state
- A penalty of **10** when falling into a hole OR the episode times out

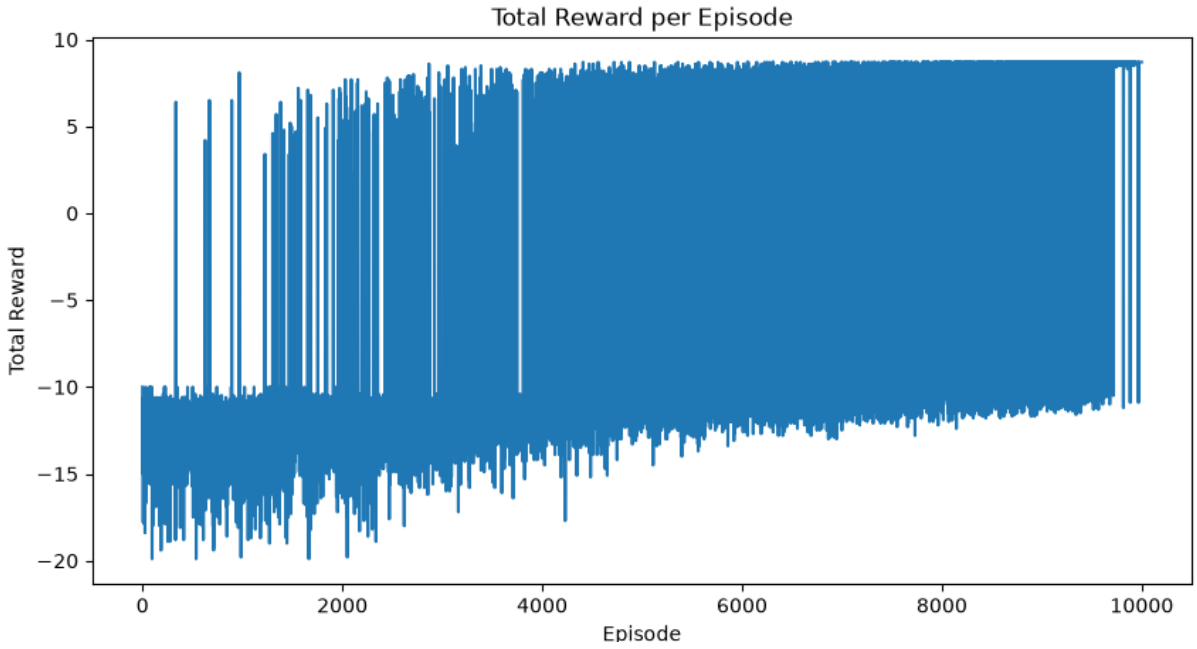
After running this program, these are the initial results:

- The average number of steps taken to reach the goal was: 18.992
- The best reward route to reach the goal was: 8.7
- The highest number of steps found to reach the goal was: 85
- A slow rise to reach the highest success rate the model could find
- There were some notable dips between the 6000 – 8000 episodes

Note: *I do understand the reward graph is rough to see, but I wanted a visual of where the program started to consistently understand a route to the end*

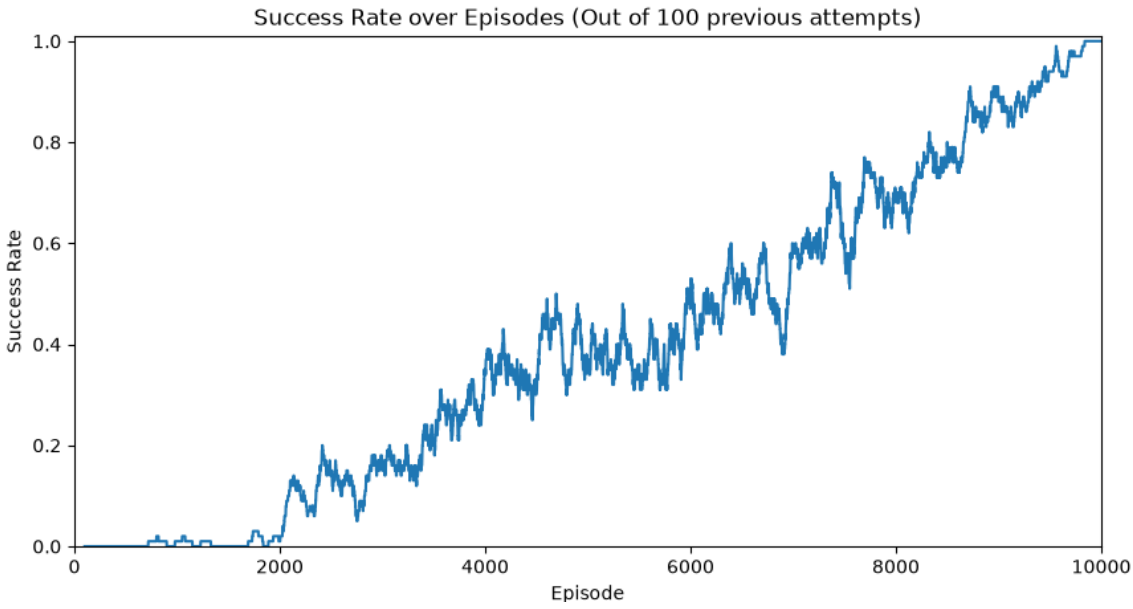
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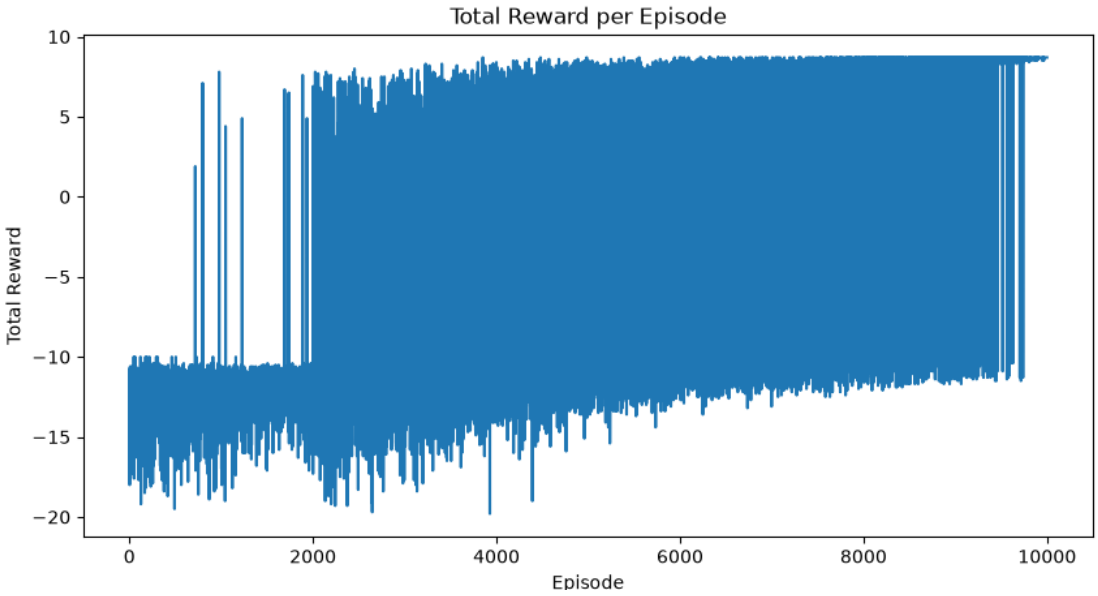




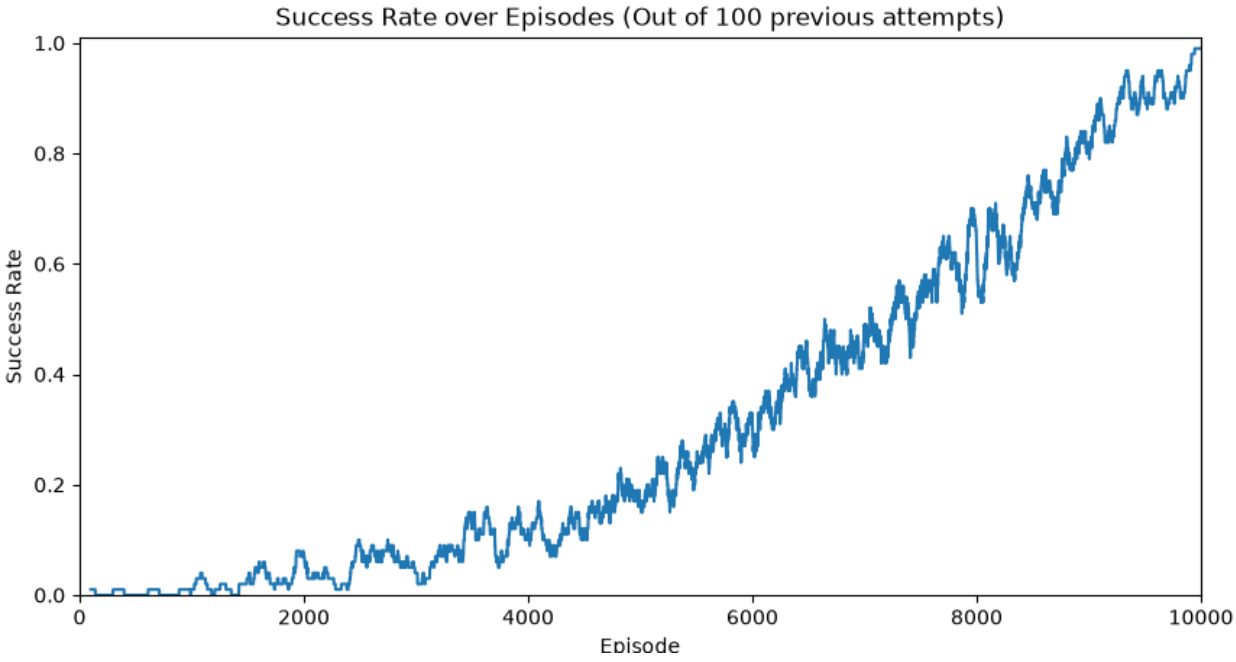
We will now be changing the learning rate using the following alpha values: 0.1 , 0.6 , 0.9

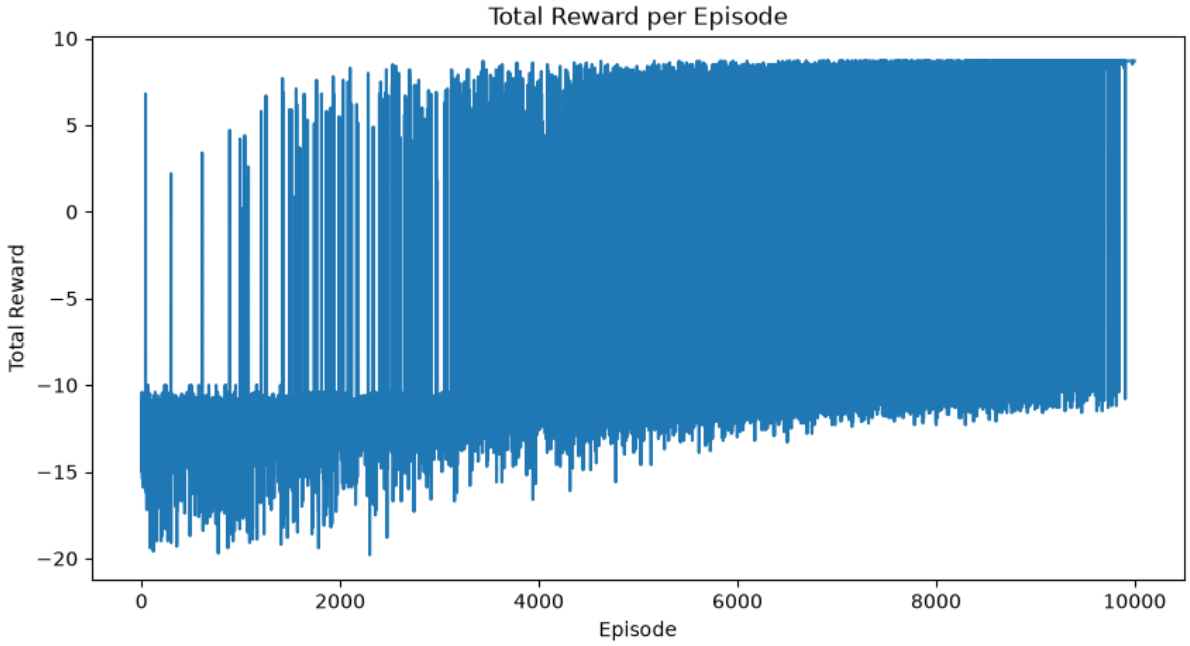
- **0.1**:
  - The average number of steps taken to reach the goal was: 21.05
  - The best reward route to reach the goal was: 8.7
  - The highest number of steps found to reach the goal was: 99
  - A much larger lull was found at the beginning of the success rate, and it only saw progress after about the 1900-episode mark.
    I could roughly draw a linear line where the training improved apart from some occasional rough dips in the graph



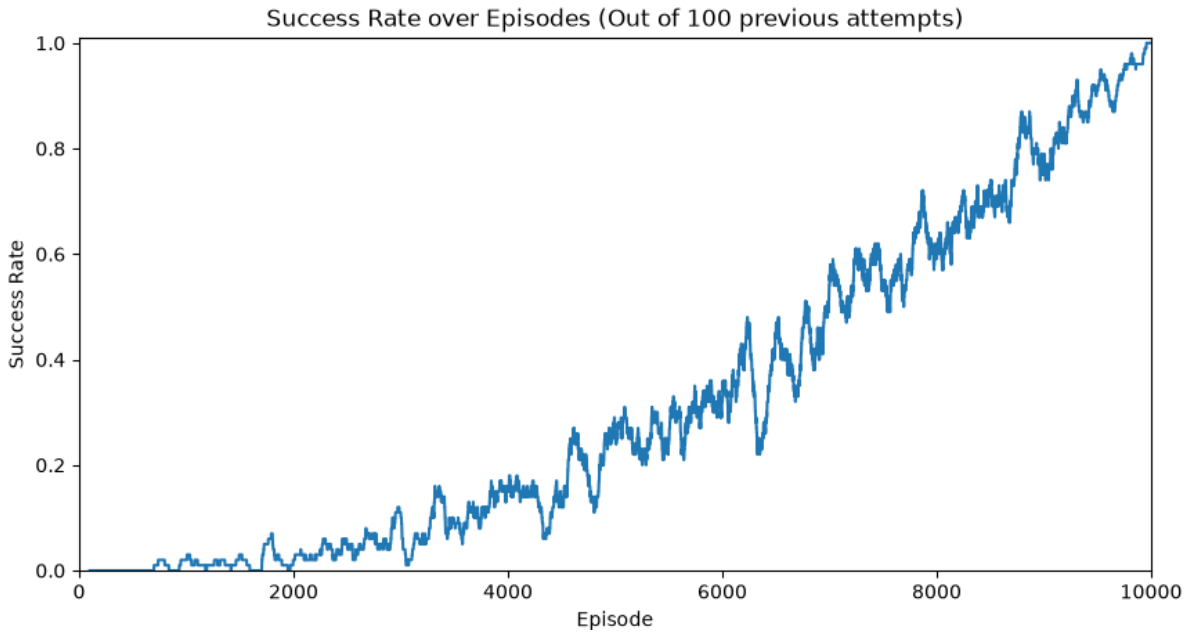


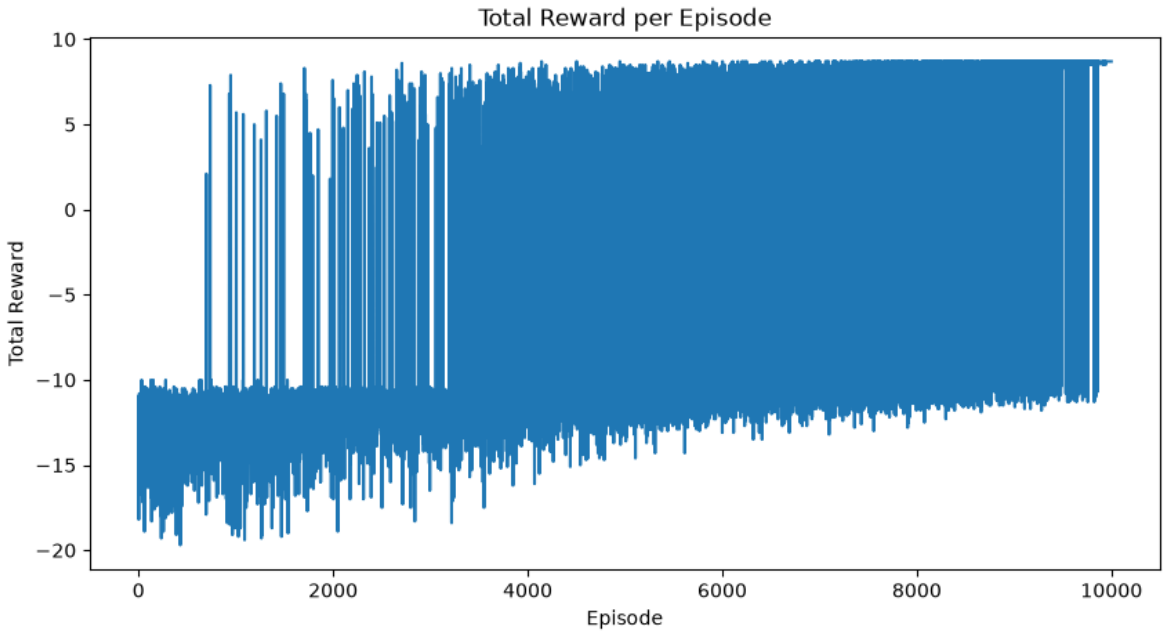
- **0.6**:
    - The average number of steps taken to reach the goal was: 18.839
    - The best reward route to reach the goal was: 8.7
    - The highest number of steps found to reach the goal was: 99
    - This found some earlier but unsustained success from the start.
      The graph shows a very slow learning rate until about the 5200-episode mark where the slope improves considerably





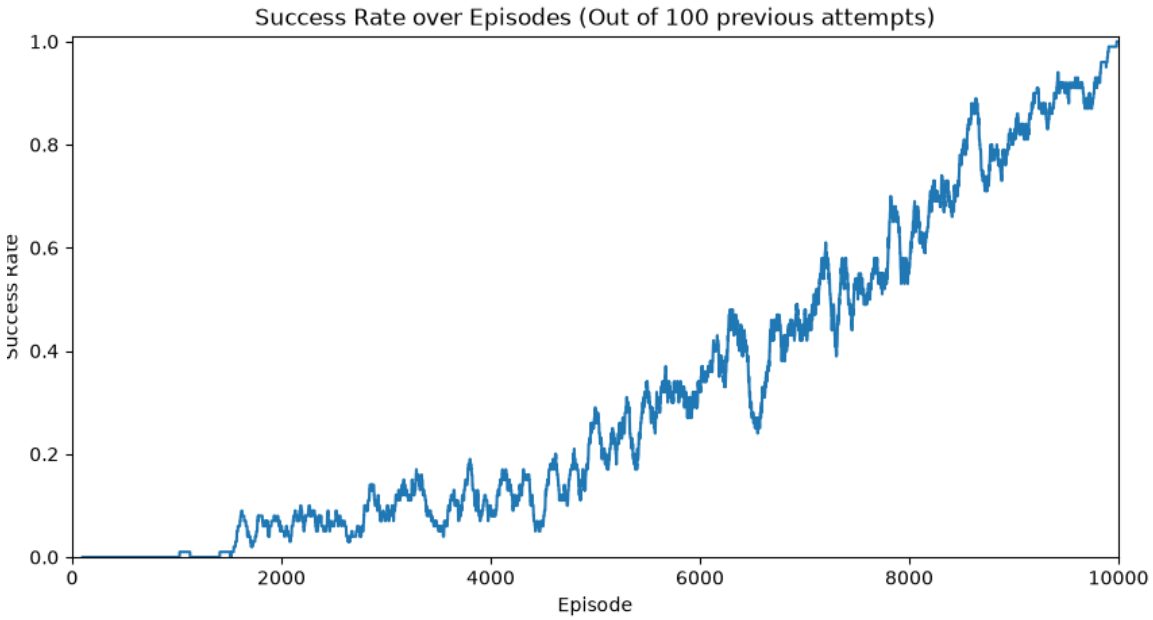
- **0.9**:
    - The average number of steps taken to reach the goal was: 18.878
    - The best reward route to reach the goal was: 8.7
    - The highest number of steps found to reach the goal was: 83
    - This had a rough learning curve at the beginning finding itself falling to nearly 0.0 rate in 100 attempts around the 3100-episode mark but eventually picked itself up.
      There is a very noticeable drop right around the 6200-episode mark then quickly recovers again after.

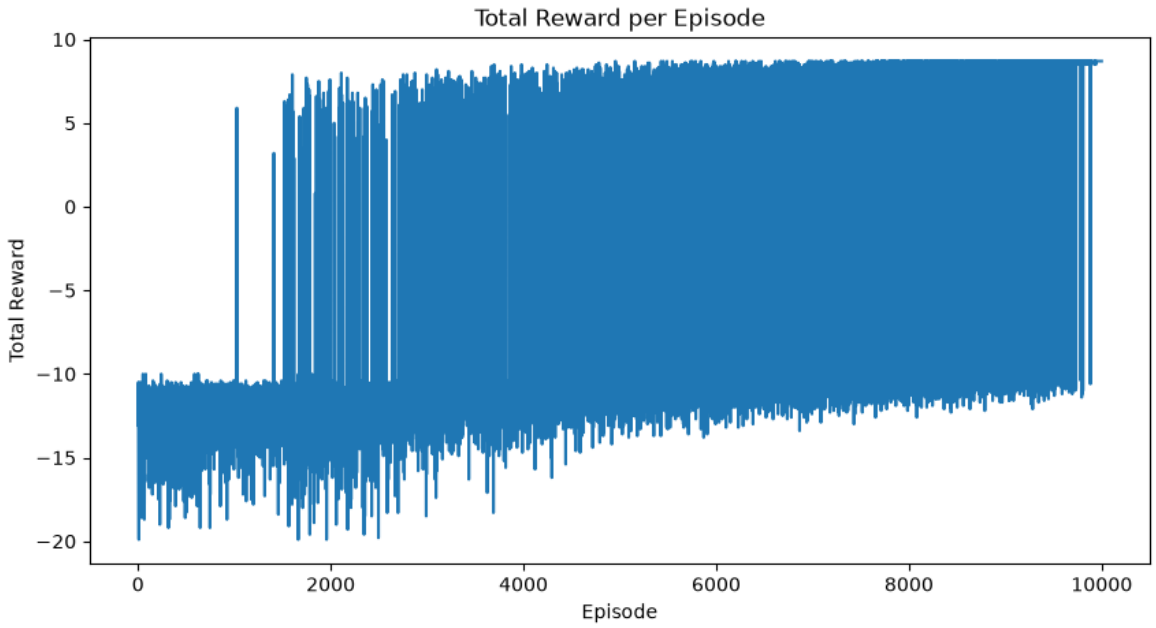




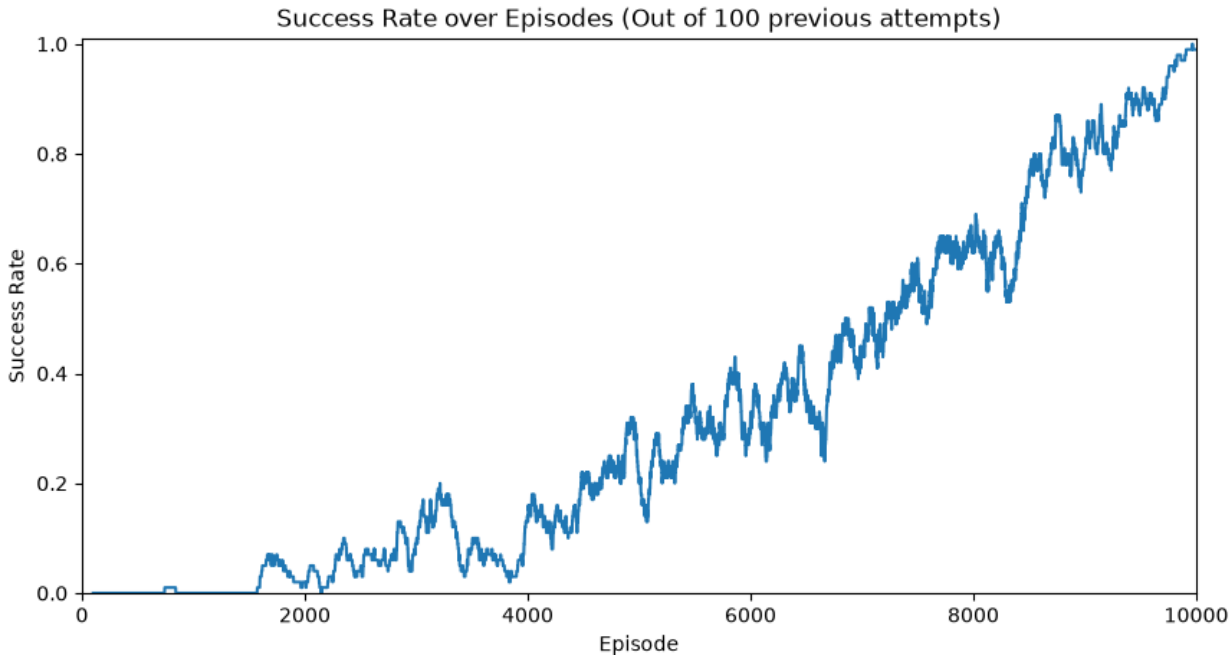
Next we will changing the discount factor using the following gamma values: 0.99 , 0.8 , 0.5

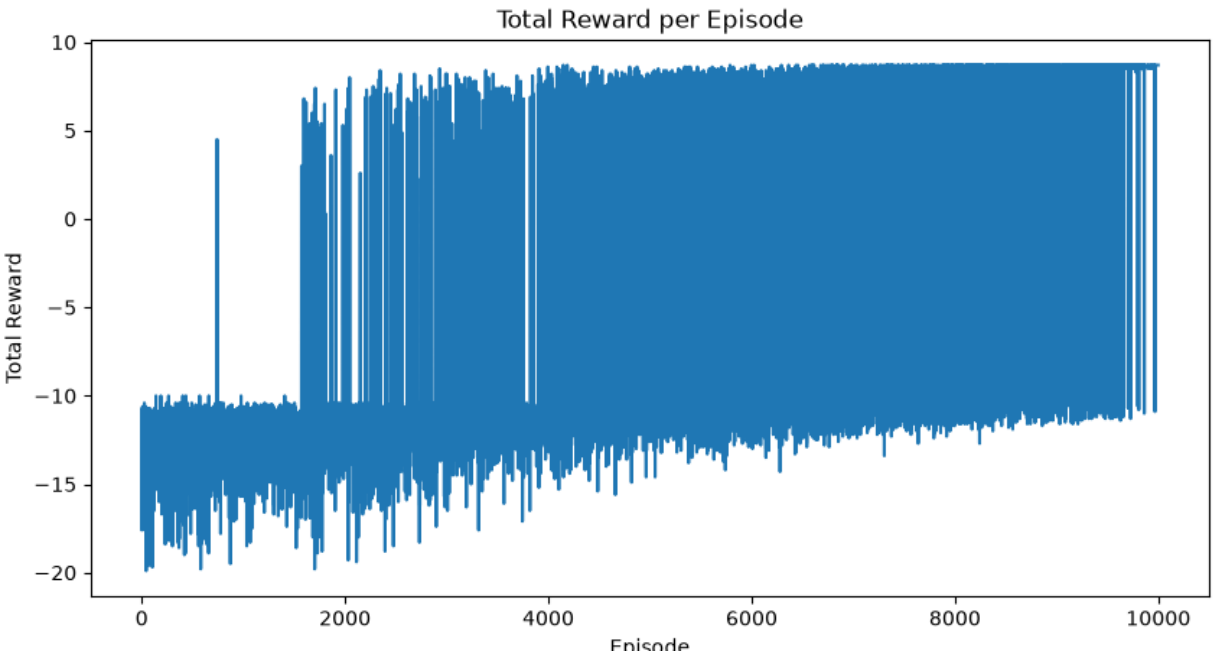
- **0.99**:
    - The average number of steps taken to reach the goal was: 19.12
    - The best reward route to reach the goal was: 8.7
    - The highest number of steps found to reach the goal was: 93
    - This had a very slow and erratic learning curve. The model kept itself under the 20% until about the 5000-episode mark. There was one large dip at about the 6200-episode mark, and a odd spike at around the 8300-episode mark then went back down following the trajectory it had before the spike.



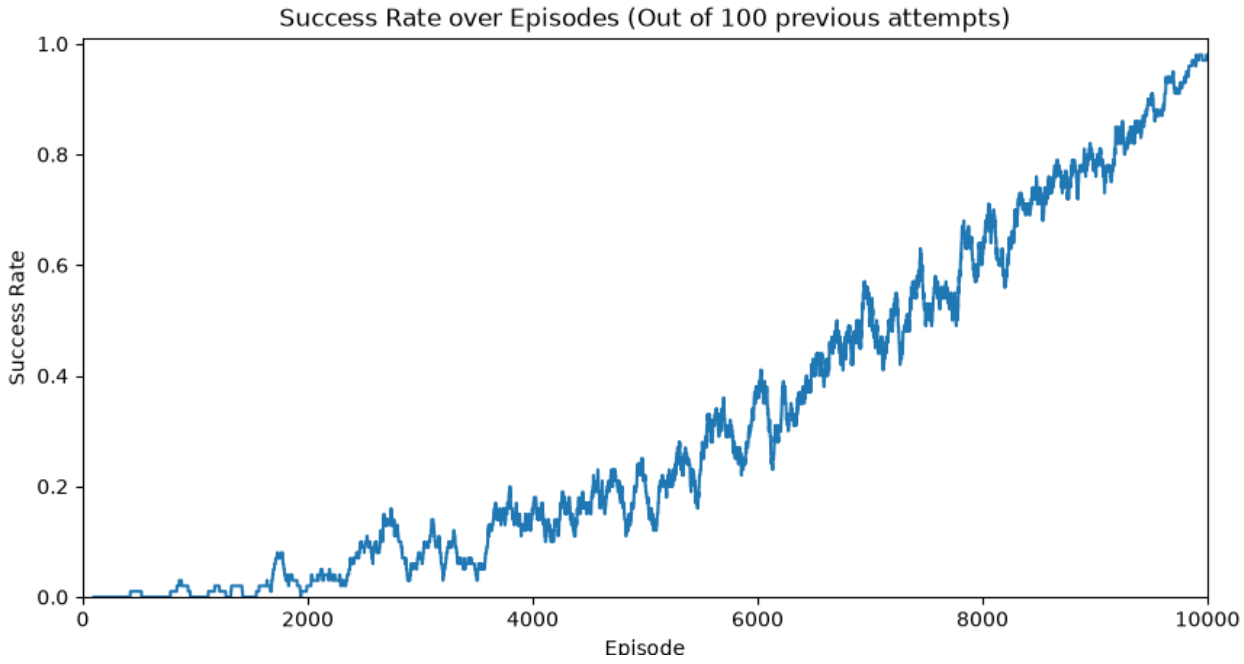


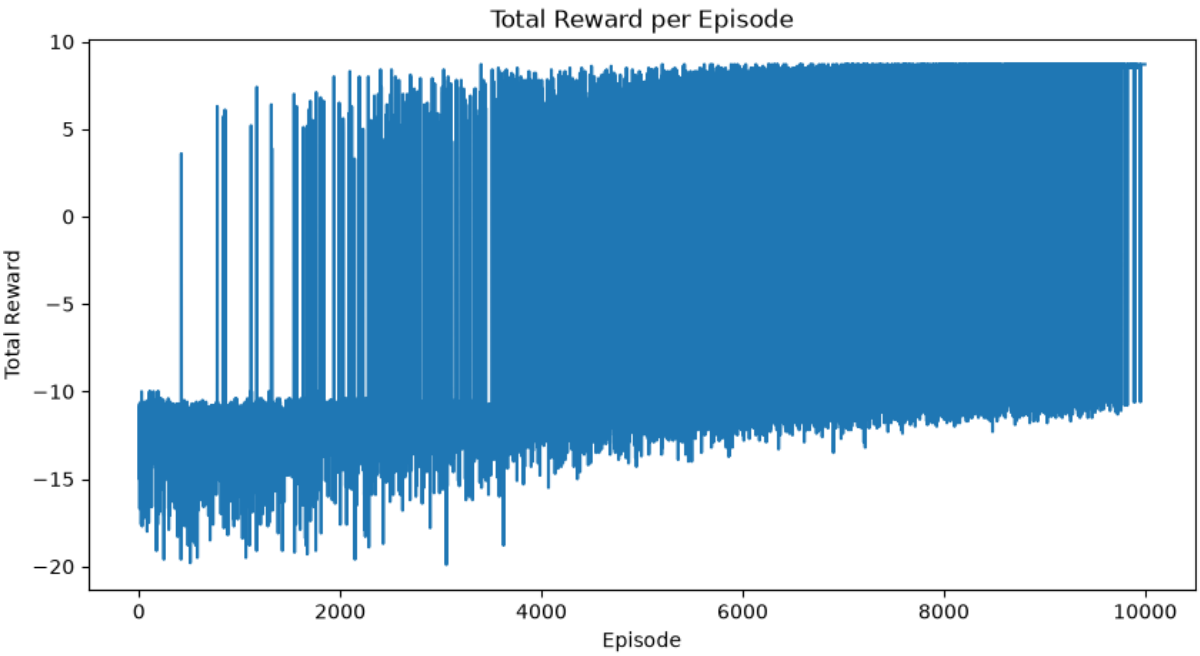
- **0.8**:
    - The average number of steps taken to reach the goal was: 18.97
    - The best reward route to reach the goal was: 8.7
    - The highest number of steps found to reach the goal was: 100
    - This had poor success until around the 1800-episode mark then could not maintain anything stable until the 4000-episode mark where it improved slowly yet erratically because of 2 notable dips then suddenly picks up around the 8200-episode mark.





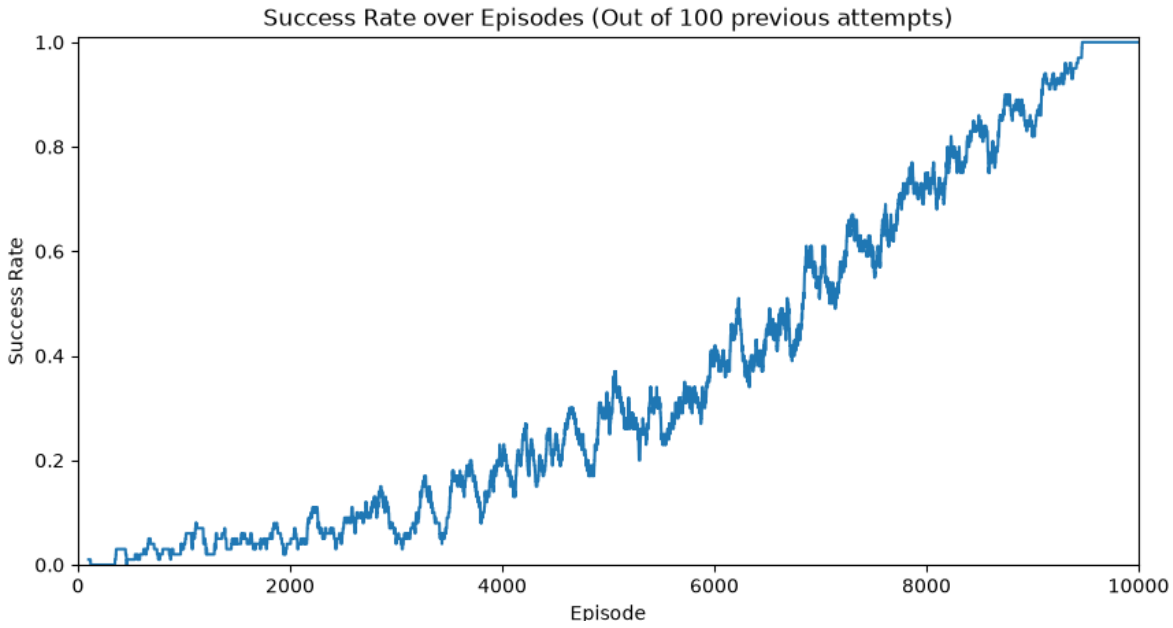
- **0.5**:
    - The average number of steps taken to reach the goal was: 18.87
    - The best reward route to reach the goal was: 8.7
    - The highest number of steps found to reach the goal was: 80
    - This had a rough start until about the 1900-episode mark, it had a rough patch until about the 3800-episode mark then had a fairly consistent rise to the top.

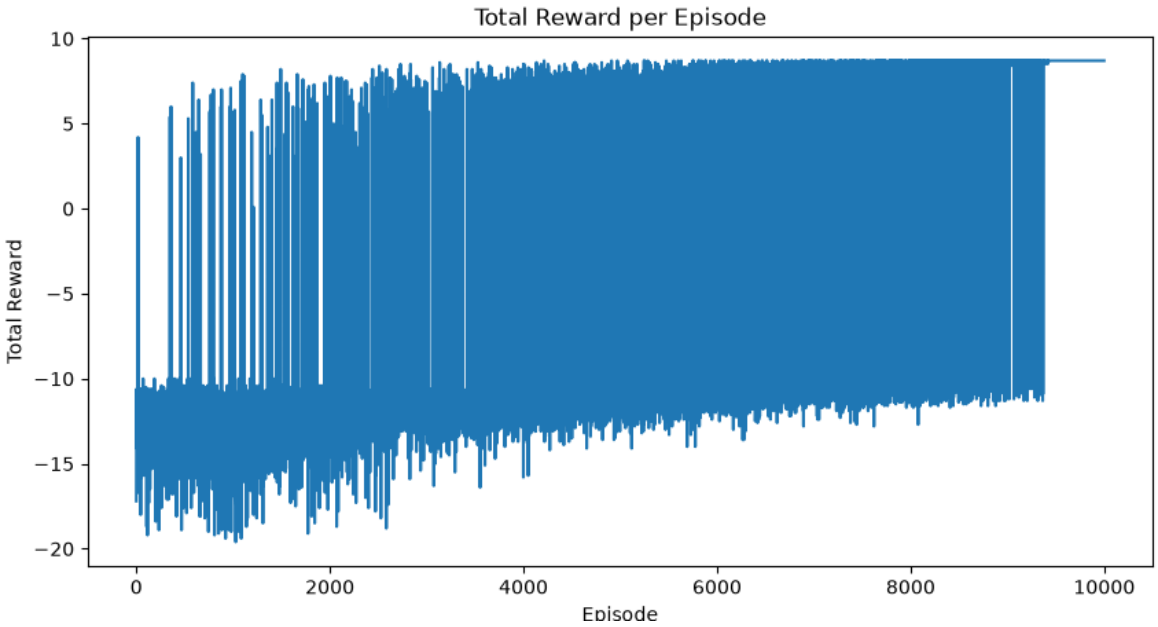




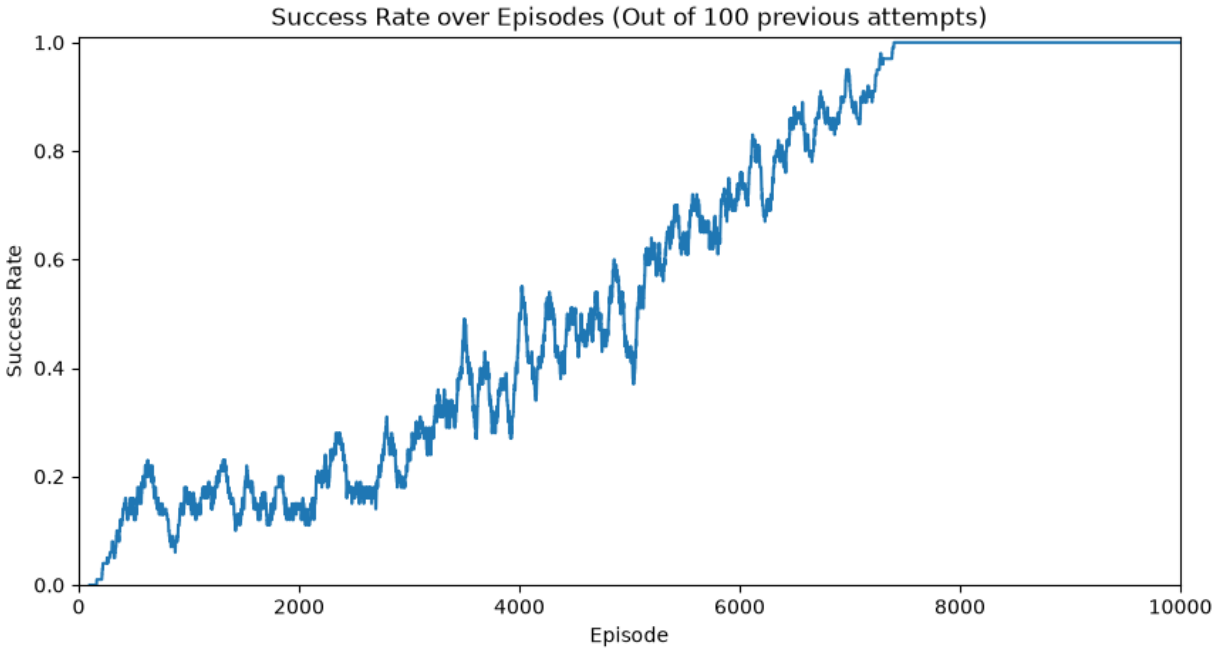
Next, we will change the epsilon value using the following values: 0.95 , 0.75 , 0.55

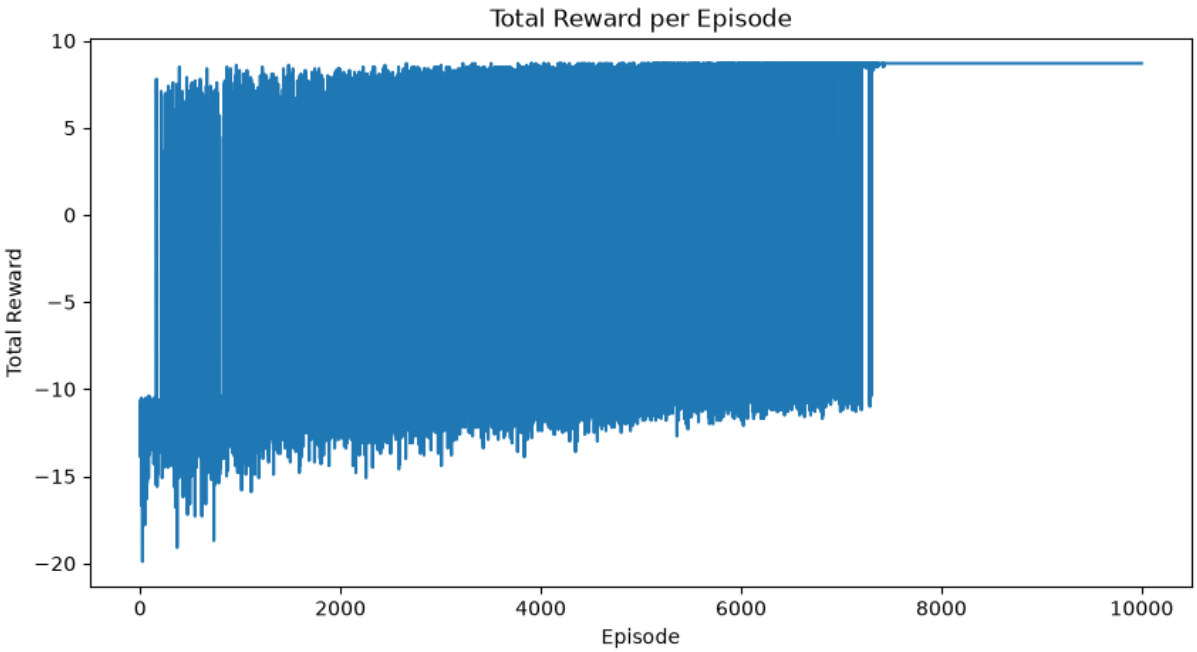
- **0.95**:
  - The average number of steps taken to reach the goal was: 18.45
  - The best reward route to reach the goal was: 8.7
  - The highest number of steps found to reach the goal was: 100
  - This had a fairly slow learning rate jumping back and forth between progress then back down to about 0.01 success rate then seemed to have picked up around the 3900-episode mark then had a fairly linear improvement after until the epsilon reached 0



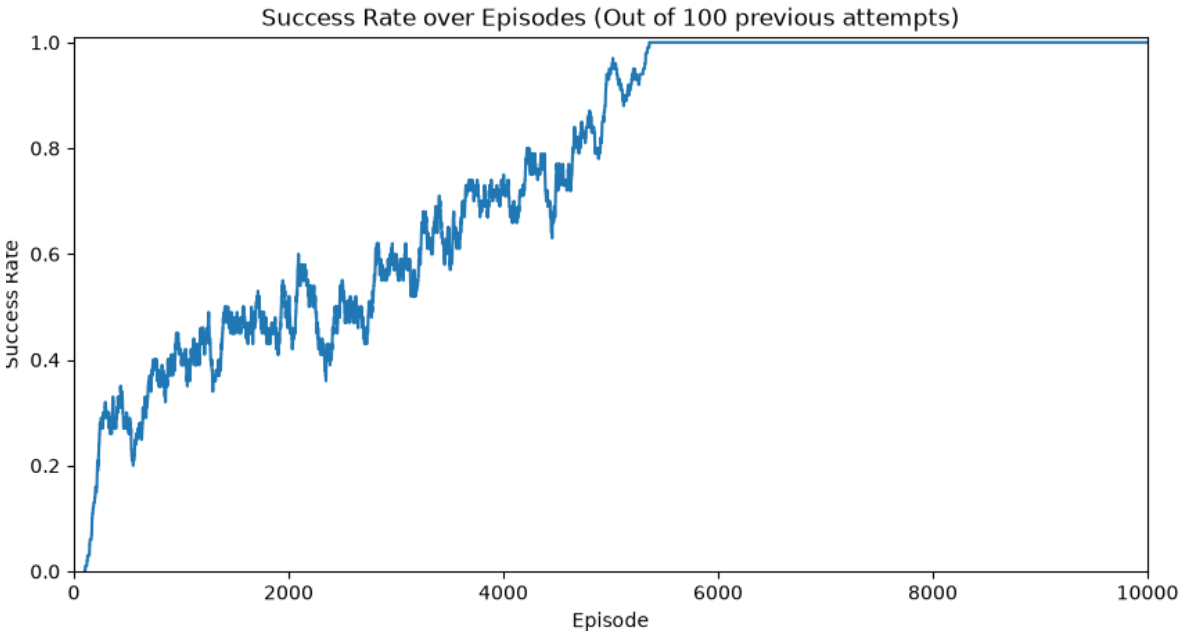


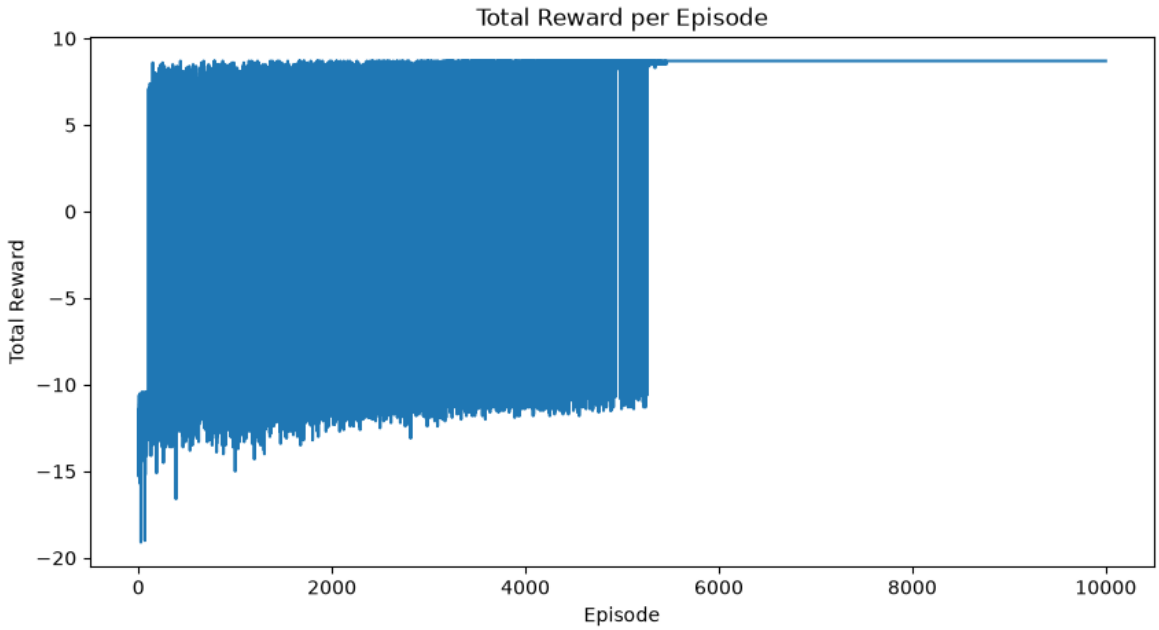
- **0.75**:
  - The average number of steps taken to reach the goal was: 16.8
  - The best reward route to reach the goal was: 8.7
  - The highest number of steps found to reach the goal was: 85
  - This had a really good success rate out the gate but stuck around the 0.17 rate until around the 2300-episode mark where it improved linearly until the epsilon reached 0





- **0.55**:
    - The average number of steps taken to reach the goal was: 15.64
    - The best reward route to reach the goal was: 8.7
    - The highest number of steps found to reach the goal was: 56
    - This had a very strong success rate improvement until it reached a 0.36 rate then it bounced around. It then very erratically improved to the 0.6 success rate and 2000-episode mark, then tanked itself. It caught itself back up at the 3000-episode mark then again slowly improved with an unstable rate until the epsilon reached 0





Out of the newly tested models, I kept the model with the following features:

- 10000 episodes
- Learning rate alpha of 0.5
- Discount factor gamma of 0.9
- Epsilon of 0.55
- Epsilon decay of 0.0001

The reason I decided this was the best performing the best because it ended with the same route distance in a substantially shorter amount of time.
Both discovered the best reward route of 8.7 while spending less time on needlessly long exploration routes requiring over 80 steps.

After running this model 30 times
Every run resulted with 14 steps and a 100% success rate along with an 8.7 mean reward.

Although the runs differed by a slightly different start route, at the end, every single run had the same meaningful results.